The main purpose of this document is to give a technical summary of existing approaches towards (offline) functional regression, and remark on their strengths/weakness. When appropriate, I will also remark on possibilities to bandit problems (mostly confidence intervals).

Let $\mathcal{X}$ be a compact Euclidean set and $\mathcal{D} = L^2(\mathcal{X})$ be the set of all square integrable functions on $\mathcal{X}$. Let $\mathbb{P}$ be the distribution on $\mathcal{D}$. The model is

$$y_i = \langle f_i, \beta_0 \rangle + \varepsilon_i$$

where $\{\varepsilon_i\}_{i=1}^n$ are centered, exogenous sub-Gaussian random variables, $\beta_0 \in \mathcal{F} \subseteq \mathcal{D}$ is an unknown regression model and $f_1, \cdots, f_N \overset{i.i.d.}{\sim} \mathbb{P}$. The inner product is taken with respect to the Lesbesgue measure on $\mathcal{X}$:

$$\langle f_i, \beta_0 \rangle := \int_{\mathcal{X}} f_i(u)\beta_0(u)\mathrm{d}u.$$

Sometimes, we will place additional assumptions on $\mathcal{F}$, or restrict the support of $\mathbb{P}$. Errors are measured in

$$\|\widehat{\beta}_n - \beta_0\|_{L^2(\mathbb{P})}^2 := \int_{\mathcal{D}} \langle f, \widehat{\beta}_n - \beta_0 \rangle^2 \mathrm{d}\mathbb{P}(f).$$

Let K be the RKHS and recall the notation that $\widetilde{f} = L_{K^{1/2}} f$, $\widetilde{\beta}_0 = L_{K^{-1/2}}\beta_0$, such that $\|\widetilde{\beta}_0\|_2 < +\infty$. Let $T = \mathbb{E}[\widetilde{f} \otimes \widetilde{f}]$ be the population covariance operator, which is decomposed as $T = \sum_{j=1}^\infty s_j \psi_j \otimes \psi_j$ for an orthonormal basis $\{\psi_j\}_{j=1}^\infty$ of $L_2(\mathcal{X})$, and $s_j \asymp j^{-2r}$.

0.1 The anisotropic KRR approach

Given observations $\{f_i, y_i\}_{i=1}^n$, let

$$T_n := \frac{1}{n}\sum_{i=1}^n \widetilde{f}_i \otimes \widetilde{f}_i$$

be the sample covariance operator, where $\widetilde{f}_i = L_{K^{1/2}} f_i$. By the spectral theorem, there exist orthonormal (w.r.t. $L^2(\mathcal{X})$) $\{\widehat{\psi}_j\}_{j=1}^n$ and $\widehat{s}_1 \geq \cdots \widehat{s}_n \geq 0$ such that

$$T_n = \sum_{j=1}^n \widehat{s}_j \widehat{\psi}_j \otimes \widehat{\psi}_j.$$

Note that T_n can only have at most n non-zero eigenvalues.

Given a function f, let $\widehat{\alpha}_j := \langle \widetilde{f}, \widehat{\psi}_j \rangle = \langle L_{K^{1/2}} f, \widehat{\psi}_j \rangle$. We then use the following anisotropic KRR estimator

$$\widehat{\beta} = (T_n + \Lambda_n(f))^{-1} \cdot \frac{1}{n}\sum_{i=1}^n y_i L_{K^{1/2}} f_i,$$

where

$$\Lambda_n(f) = \sum_{j=1}^n \lambda_j(f) \widehat{\psi}_j \otimes \widehat{\psi}_j; \qquad \lambda_j(f) = \begin{cases} \widehat{s}_j \sqrt{\lambda_0}/|\widehat{\alpha}_j|, & j \leq J_1; \\ +\infty, & j > J_1, \end{cases}$$

where $J_1 \leq n$ is a cut-off threshold to be determined later. Note that this is a truncation + KRR approach. For notational convenience, we will drop the dependency on f whenever there is no ambiguity. An estimate of $\langle f, \beta_0 \rangle$ is the obtained as $\langle L_{K^{1/2}} f, \widehat{\beta} \rangle$.

0.2 Analysis of bias

Recall the definition that $\widetilde{\beta}_0 = L_{K^{-1/2}}\beta_0$. Let

$$\bar{\beta} := \mathbb{E}[\widehat{\beta}] = (T_n + \Lambda_n)^{-1}\frac{1}{n}\sum_{i=1}^{n}\langle \widetilde{f}_i, \widetilde{\beta}_0\rangle \widetilde{f}_i = (T_n + \Lambda_n)^{-1}T_n\widetilde{\beta}_0.$$

Let $\widehat{\theta}_j := \langle \widetilde{\beta}_0, \widehat{\psi}_j\rangle$. Then

$$\widetilde{\beta}_0 - \bar{\beta} = (T_n + \Lambda_n)^{-1}\Lambda_n\widetilde{\beta}_0 = \sum_{j=1}^{n}\frac{\lambda_j\widehat{\theta}_j}{\widehat{s}_j + \lambda_j}\widehat{\psi}_j.$$

Subsequently, the bias is upper bounded by

$$\left|\langle \widetilde{f}, \widetilde{\beta}_0 - \bar{\beta}\rangle\right| = \left|\sum_{j=1}^{n}\frac{\lambda_j\widehat{\theta}_j\widehat{\alpha}_j}{\widehat{s}_j + \lambda_j}\right|.$$

From the definition of λ_j, it holds for every $j \in [n]$ that

$$\frac{\lambda_j|\widehat{\theta}_j\widehat{\alpha}_j|}{\widehat{s}_j + \lambda_j} \le \begin{cases} \sqrt{\lambda_0}|\widehat{\theta}_j|, & j \le J_1; \\ |\widehat{\theta}_j\widehat{\alpha}_j|, & j > J_1. \end{cases}$$

Let $J_1 := n^{4r/(4r^2-1)} = J_0^{2r/(2r-1)}$ where $J_0 = n^{1/(2r+1)} = \lambda_0^{-1/(2r)}$. We use the following approach to upper bound the RHS of the above inequality:

$$\begin{aligned}\left|\sum_{j=1}^{n}\frac{\lambda_j\widehat{\theta}_j\widehat{\alpha}_j}{\widehat{s}_j + \lambda_j}\right| &\le \sqrt{\lambda_0}\left(\sum_{j=1}^{J_1}|\widehat{\theta}_j|\right) + \sqrt{\sum_{j=J_1+1}^{n}\widehat{\theta}_j^2}\sqrt{\sum_{j=J_1+1}^{n}|\widehat{\alpha}_j|^2} \\ &\le \sqrt{\lambda_0}\|\widehat{\theta}_{J_1}\|_1 + \lambda_0\|\widehat{\theta}\|_2\|\widehat{\alpha}_{(J_1+1):n}\|_2.\end{aligned}$$

It should also be noted that, if $\sum_{j=J_1+1}^{n}\widehat{s}_j \lesssim \sum_{j=J_1+1}^{n}s_j$, then

$$\mathbb{E}[\|\widehat{\alpha}_{(J_1+1):n}\|_2^2] \lesssim \int_{J_1}^{\infty}t^{-2r}\mathrm{d}t \lesssim J_1^{1-2r} = J_0^{-2r} = \lambda_0.$$

0.3 Analysis of perturbation

We impose the following two main assumptions, on "population" coefficients $\theta_j := \langle \widetilde{\beta}_0, \psi_j\rangle$:

$$\sum_{j=1}^{\infty}\theta_j^2 < +\infty, \qquad \sum_{j=1}^{\infty}|\theta_j| < +\infty.$$

The first condition is standard which simply means $\|\widetilde{\beta}_0\|_2 < +\infty$. This condition is also basis-invariance so applicable to $\{\widehat{\theta}_j\}_j$ too. The main issue is the second assumption, which is *not* basis invariant.

We state the following result, which controls the behaviors of $\{\widehat{\theta}_j\}$ and $\{\widehat{s}_j\}$.

Lemma 0.1. *Suppose the following additional conditions hold for every $j \in [n]$:*

1. $s_j \asymp j^{-2r}$ *and* $s_j - s_{j+1} \gtrsim j^{-2r-1}$;

2. *The random variable* $\widehat{\alpha}_j = \langle \widetilde{f}, \widehat{\psi}_j \rangle$, *with randomness on* $\widetilde{f}$ *only, is sub-Gaussian with variance proxy* s_j.

Then, if $r > 1 + \sqrt{5}/2 = 2.118 =: r_0$, *the following holds with probability* $1 - o(1)$*:*

1. $\sum_{j=1}^{J_1} |\widehat{\theta}_j| \le \sum_{j=1}^{J_1} |\theta_j| + o(1)$;
2. $\widehat{s}_j \gtrsim j^{-2r}$ *for all* $1 \le j \le J_1$;
3. $\sum_{j=J_1+1}^{n} \widehat{s}_j \lesssim J_1^{-2r+1}$.

Remark 0.1. The sub-Gaussianity assumption and the spectral gap assumption can be somewhat relaxed, at the cost of a larger value of s_0. It should be noted that both assumptions are adopted for fPCA. We do *not* adopt the fPCA assumption of decays on $\{\theta_j\}$, however, because that leads to significant sub-optimality.

Proof of Lemma 0.1. Express both T and T_n on the first n eigenfunctions $\{\psi_i\}_{i=1}^n$ of T. Then $T = \mathrm{diag}(s_1, \cdots, s_n)$ with the standard basis as its eigenvectors, and T_n is a PSD matrix. We will carry out our proof in this $n \times n$ finite matrix field to simplify matters.

We will use the following inequality as the key workhorse of this proof: with probability $1 - o(1)$ uniformly for every $i, j \in [n]$,

$$\big|[T_n]_{ij} - T_{ij}\big| \lesssim \frac{i^{-r} j^{-r}}{\sqrt{n}}.$$

We can potentially tolerate higher-order terms in this inequality, if we want to relax the sub-Gaussianity assumption, but Bernstein with the simple $|T_{ij}| \le 1$ is *not* enough. The rest of the proof are conditioned on the success event that the above holds.

Consider the submatrix of T_n corresponding to $\psi_{J_1+1}, \cdots, \psi_n$. The trace formula implies that

$$\sum_{j=J_1+1}^{n} \widehat{s}_j \le \sum_{j=J_1+1}^{n} [T_n]_{jj} \lesssim \sum_{j=J_1+1}^{n} j^{-2r} + O\left(\frac{j^{-2r}}{\sqrt{n}}\right) \lesssim J_1^{-2r+1},$$

which proves the third property.

Now, let A and A_n be the $J_1 \times J_1$ sub-matrices of T and T_n corresponding to the eigenspace spanned by $(\psi_1, \cdots, \psi_{J_1})$. We will prove that, for every $j \in [J]$, there exists $w \in \mathbb{R}^d$ satisfying $\|w\|_2 = o(J_1^{-1})$ such that

$$A_n(e_j + w) = s_j(e_j + w). \tag{1}$$

This shows that $\widehat{s}_j = s_j(1 + o(J_1^{-1}))$. Because the eigengaps of the first J_1 eigenfunctions are larger than J_1^{-1}, this ensures that the top-J_1 eigenfunctions of T and T_n have the same span $\mathrm{span}(\psi_1, \cdots, \psi_{J_1})$, and furthermore $\|\widehat{\psi}_j - \psi_j\|_2 = o(J_1^{-1})$. This is sufficient to prove the first two properties.

Let $H = A_n - A$ and consider $w_j = 0$ which is simply a Gauge choice. Because $Ae_j = s_j e_j$, Eq. (1) implies

$$(A - s_j I + H)w = -He_j.$$

Let $\bar{H} \in \mathbb{R}^{(J_1-1)\times(J_1-1)}$ be the submatrix formed by removing the jth row and jth column of H. Let $h \in \mathbb{R}^{J_1-1}$ be the jth column of H, removing H_{jj}. Let $D = \mathrm{diag}(s_1 - s_j, \cdots, s_{j-1} - s_j, s_{j+1} - s_j, \cdots, s_{J_1} - s_j) \in \mathbb{R}^{(J_1-1)\times(J_1-1)}$. Let $\bar{w} = (w_1, \cdots, w_{j-1}, w_{j+1}, \cdots, w_n) \in \mathbb{R}^{J_1-1}$. Then

$$\bar{w} = -(D + \bar{H})^{-1} h = -D^{-1/2}(I + \widetilde{H})^{-1} D^{-1/2} h,$$

where $\widetilde{H} = D^{-1/2} \bar{H} D^{-1/2}$.

For any $i, k \neq j$, we have that

$$\left|\widetilde{H}_{ik}\right| \lesssim \frac{1}{\sqrt{n}} \frac{i^{-r}k^{-r}}{\sqrt{|(s_i - s_j)(s_k - s_j)|}} =: \frac{1}{\sqrt{n}} z_i z_k \qquad \text{where } \; z_k = \frac{k^{-r}}{\sqrt{|s_k - s_j|}}.$$

In a similar way, $D^{-1/2}h$ can also be upper bounded component-wise as

$$\left|[D^{-1/2}h]_k\right| \lesssim \frac{1}{\sqrt{n}} \frac{j^{-r}k^{-r}}{\sqrt{|s_k - s_j|}} = \frac{j^{-r}}{\sqrt{n}} z.$$

Let $C > 0$ be a sufficiently large numerical constant (potentially depending on constants in spectrum decay and spectral gaps). Discuss three cases:

1. The case of $k \leq j/C$. In this case $|s_k - s_j| \gtrsim s_k \gtrsim k^{-2r}$, and therefore $|z_k| \lesssim O(1)$.
2. The case of $k \geq Cj$. In this case $|s_k - s_j| \gtrsim s_j \gtrsim j^{-2r}$, and therefore $|z_k| \lesssim O(j^r k^{-r})$.
3. The case of $j/C \leq k \leq Cj$. In this case $|s_k - s_j| \gtrsim |k - j| j^{-2r-1}$, and therefore $|z_k| \lesssim O(\sqrt{j/|k-j|})$.

Because $\widetilde{H} = zz^\top/\sqrt{n}$, summarizing the above cases we obtain

$$\|\widetilde{H}\|_{\mathrm{op}} \lesssim \frac{1}{\sqrt{n}}\|z\|_2^2 = \frac{1}{n}\sum_{k \neq j} |z_k|^2 \lesssim \frac{1}{\sqrt{n}}\left(j + j^{2r}\int_j^\infty t^{-2r}\mathrm{d}t + j\int_1^{Cj}\frac{\mathrm{d}t}{t}\right) \lesssim \frac{j\ln(j+1)}{\sqrt{n}}. \tag{2}$$

With $r > 1 + \sqrt{2}/2 = 1.207$, we have $J_1 = n^a$ for some $a < 1/2$. Therefore, $\|\widetilde{H}\|_{\mathrm{op}} = o(1)$. [Remark: utilizing the fact that $\mathbb{E}[\widetilde{H}_{jk}] = 0$ we might get tighter control of $\|\widetilde{H}\|_{\mathrm{op}}$ with high probability. Highly likely $\sqrt{j/n}$. However, it is going to be harder to bound $\widetilde{H}^\ell$ later since we don't have element-wise non-negativity.]

We can then do the following Taylor expansion using $(I + \widetilde{H})^{-1} = \sum_{\ell=0}^\infty (-1)^\ell \widetilde{H}^\ell$:

$$\begin{aligned}
\|w\|_2 &\leq \sum_{\ell=0}^\infty \|D^{-1/2}\widetilde{H}^\ell D^{-1/2}h\|_2 \leq \sum_{\ell=0}^\infty \frac{j^{-r}}{\sqrt{n}}\left\|D^{-1/2}\left(\frac{1}{\sqrt{n}}zz^\top\right)^\ell z\right\|_2 \\
&\leq \sum_{\ell=0}^\infty \frac{j^{-r}}{\sqrt{n}}\left(\frac{j}{\sqrt{n}}\right)^\ell \|D^{-1/2}z\|_2 \lesssim \frac{j^{-r}}{\sqrt{n}}\|D^{-1/2}z\|_2 \\
&\lesssim \frac{j^{-r}}{\sqrt{n}}\left(\int_1^j \frac{\mathrm{d}t}{t^{-2r}} + j^{2r}\int_j^{J_1}\frac{t^{-2r}}{j^{-2r}}\mathrm{d}t + j\cdot\int_1^{Cj}\frac{1}{tj^{-2r-1}}\frac{\mathrm{d}t}{t}\right)^{1/2} \\
&\lesssim \frac{j^{-r}}{\sqrt{n}}\left(j^{2r+1} + j^{2r+1} + j^{2r+2}\right)^{1/2} = \frac{j}{\sqrt{n}}.
\end{aligned}$$

With $4r^2 - 8r - 1 > 0$, implied by $r > 2 + \sqrt{5}/2 = 2.118$, $j/\sqrt{n} < 1/J_1$ and this proves the desired results. □

References